COURSE 2 · SECTION 2 · WORKED SOLUTION

Which seat ring revision was fitted?

Section 2 · Products, parts, BOM and revisions

DESIGN QUESTION

Which revision of the P300 valve seat ring was in effect in each build month, does that match what the Builds sheet says was fitted, and how many seat rings did Engxtech fit in total?

Steps

1. Open Section03_Start.pbix (or Engxtech_S02.pbix) > File > Save as > Engxtech_S02_Exercise.
2. Home > Excel workbook > course workbook > tick Builds only > Transform data. BuildMonth = Date, UnitsBuilt = Whole number. Close & Apply.
3. Check the key: Table view > Revisions > select PartRev; the status bar shows 16 distinct values in 16 rows.
4. Model view: drag Products[ProductID] onto Builds[ProductID]; one-to-many, single direction.
5. Table view > Builds > New column: RevByDate (below). Then New column: RevCheck.
6. Report view: table with Builds[BuildMonth], Builds[ProductID], SeatRingRev, RevByDate, RevCheck; and a card with Count of RevCheck filtered to Match.
7. New measures P300 Rev On Date and Seat Rings Fitted (on Builds) on cards.

DAX calculated columns (Builds)

```
RevByDate =
VAR d = Builds[BuildMonth]
VAR lastRel =
    MAXX (
        FILTER ( Revisions, Revisions[PartID] = "P300" && Revisions[ReleaseDate] <= d ),
        Revisions[ReleaseDate]
    )
RETURN
    LOOKUPVALUE ( Revisions[Rev], Revisions[PartID], "P300", Revisions[ReleaseDate], lastRel )

RevCheck = IF ( Builds[RevByDate] = Builds[SeatRingRev], "Match", "Different" )
```

DAX measures

```
P300 Rev On Date =
VAR d = DATE ( 2026, 1, 26 )        -- then try DATE ( 2026, 9, 30 )
VAR lastRel =
    MAXX (
        FILTER ( ALL ( Revisions ), Revisions[PartID] = "P300" && Revisions[ReleaseDate] <= d ),
        Revisions[ReleaseDate]
    )
RETURN
    LOOKUPVALUE ( Revisions[Rev], Revisions[PartID], "P300", Revisions[ReleaseDate], lastRel )

Seat Rings Fitted =
SUMX (
    Builds,
    Builds[UnitsBuilt]
        * LOOKUPVALUE ( BOM[QtyPer], BOM[ProductID], Builds[ProductID], BOM[PartID], "P300" )
)
```

Why 3 rows differ: BuildMonth is stored as the 1st of the month. P300-B was released on 12 May 2025, so on 1 May 2025 the date lookup still says rev A, while the Builds sheet records that rev B seat rings were fitted that month. A date lookup on a monthly table cannot see a mid-month cut-in; the fitted rev (SeatRingRev) is the field to trust, and the lookup is a check on it.

Seat rings vs product units: CP-100 and BP-200 use 2 seat rings each, MV-300 uses 1. Keep both numbers apart; the field ppm in Section 9 is per product unit, not per seat ring.

Expected result

Check	Result
Rows in Revisions (unique PartRev keys)	**16**
P300 revisions	**3**
P300 rev in effect on 26 Jan 2026	**B**
P300 rev in effect on 30 Sep 2026	**C**
Builds rows where RevByDate = SeatRingRev	**53 of 56**
Different rows	**3 rows, May 2025**
Product units built	**40,962**
Seat rings fitted	**73,503**
Seat rings in one of each product	**5**